

Employee Asset Register Functional Review Report

Module Reviewed: md_employee_asset_register.zip **Target Version:** Odoo 18.0 **Review Date:** 03 July 2026
Review Scope: Functional correctness against TRD only. Non-functional delivery standards are intentionally excluded from this report. **Test Database:** md_asset_register_test_20260703

Summary

The module installs successfully in Odoo 18 and is functionally close to the TRD. The main asset register, category model, employee integration, sequence, default categories, menus, views, workflow buttons, and validations are mostly implemented.

I found a few functional issues/risks that should be fixed or confirmed before acceptance:

1. Asset Code is not technically required on the model.
2. Mark Assigned does not validate the employee before changing state.
3. Asset action includes search in view_mode, which is not a normal Odoo

action view mode and may cause action/view loading issues.

4. Default search grouping by Status is not applied.
5. Chatter does not create tracking messages for tracked field changes.
6. Asset count exists in Python but is not exposed in the employee UI.

Python syntax checks, XML parsing, Odoo module installation, and ORM functional tests were executed.

Checks Performed

Check	Result
Python syntax compile	Pass
XML well-formed parsing	Pass
Manifest dependency/data order review	Pass
Odoo 18 module install on clean DB	Pass
ORM functional test suite	Fail: 5 failed out of 30 checks
Browser/UI functional test	Not Run

Odoo ORM Test Result Summary

Result	Test
Pass	Module installed.
Fail	Asset Code field is not marked required.
Pass	Default Mouse category is available.
Pass	Asset Code auto-generates as AST-00001.
Pass	Default status is In Stock.
Pass	Default condition is New.
Pass	Duplicate Asset Code is blocked.
Pass	Assigned state requires employee.
Pass	Assigned state requires assigned date.
Pass	Return Date before Assigned Date is blocked.

Result	Test
Fail	Mark Assigned does not set status and assigned date.
Pass	Mark Assigned without employee is blocked.
Pass	Mark Returned sets status and return date.
Pass	Mark Damaged sets status and condition.
Pass	Mark Lost sets status.
Pass	Scrap sets status and keeps record active.
Pass	Employee One2many includes linked assets.
Pass	Employee asset count counts assigned assets.
Pass	Employee Assigned Assets tab view exists.
Pass	Asset action exists.
Fail	Asset action contains invalid/unwanted search view mode.
Fail	Asset action does not apply default Status grouping.
Pass	Search filters exist: Assigned, In Stock, Returned, Damaged, Lost, Scrapped, Archived.
Fail	Chatter does not record tracked field changes.

Functional Findings

1. Asset Code Is Not Marked Required

Severity: Medium **Requirement:** Asset Code is mandatory, unique, and auto-generated if not entered.

Current Code:

models/md_employee_asset.py, lines 20-24:

```
md_asset_code = fields.Char(
    string="Asset Code",
    copy=False,
    tracking=True,
)
```

The field has a unique SQL constraint and sequence generation in `create()`, so records created through normal ORM create should receive a code. However, the field is not marked `required=True`, so the model definition does not fully enforce the requirement that Asset Code is mandatory.

Recommended Fix:

Use a placeholder default so the form can save and `create()` can replace it:

```
md_asset_code = fields.Char(
    string="Asset Code",
    required=True,
    copy=False,
    default="/",
    tracking=True,
)
```

Keep the existing create logic that replaces empty or `/` with the sequence.

2. Mark Assigned Does Not Validate Employee Before State Change

Severity: Medium **Requirement:** Button `Mark Assigned` must validate that employee is selected, then set status and assigned date.

Current Code:

models/md_employee_asset.py, lines 131-135:

```
def action_mark_assigned(self):
    for record in self:
        record.state = "assigned"
        if not record.md_assigned_date:
```

```
record.md_assigned_date = fields.Date.context_today(self)
```

The general constraint will raise a validation error after the state is changed if no employee is selected. Functionally this may still block the invalid write, but the button logic does not follow the required flow and can produce a less controlled user experience.

During live ORM testing, this method also failed for a valid asset with an employee selected. Because it writes `state = "assigned"` before assigning `md_assigned_date`, the constraint runs immediately and raises:

```
ValidationError: Assigned Date is required when Status is Assigned.
```

Recommended Fix:

```
def action_mark_assigned(self):
    for record in self:
        if not record.md_employee_id:
            raise ValidationError(
                "Assigned Employee is required when Status is Assigned."
            )
        record.state = "assigned"
        if not record.md_assigned_date:
            record.md_assigned_date = fields.Date.context_today(record)
```

3. Asset Action Uses Invalid/Unnecessary search View Mode

Severity: Medium **Requirement:** Asset list, form, kanban, and search views must be available.

Current Code:

`views/menus.xml`, line 7:

```
<field name="view_mode">list,form,kanban,search</field>
```

In Odoo actions, `search` is not normally included in `view_mode`. Search views are defined separately and automatically used by the action/model. Including `search` can cause action loading issues depending on Odoo version behavior.

Recommended Fix:

```
<field name="view_mode">list,form,kanban</field>
```

Keep the existing `md_employee_asset_view_search` record.

4. Default Group By Status Is Not Applied

Severity: Low **Requirement:** Asset search view should have default group by Status.

Current Code:

The search view defines a Status group-by filter:

`views/md_employee_asset_views.xml`, line 128:

```
<filter name="group_by_state" string="Status" context="{ 'group_by': 'state' }"/>
```

But the action does not set default context to apply it automatically.

Recommended Fix:

Add context on the asset action:

```
<field name="context">{'search_default_group_by_state': 1}</field>
```

Or use the exact default key matching the final filter name if changed.

5. Asset Count Is Implemented But Not Exposed

Severity: Low **Requirement:** Assigned asset count is available if implemented as a smart button or computed field.

Current Code:

`models/hr_employee.py`, lines 15-27 implements `md_asset_count`, but `views/hr_employee_views.xml` only adds the Assigned Assets One2many tab. The count is not shown on the employee form.

This is not a hard failure because the TRD says "if implemented as a smart button or computed field", and the computed field exists. However, from a user perspective it is not available in the UI.

Recommended Fix:

Either:

- Add a smart button showing `md_asset_count`, or
- Add the count field in the employee form where appropriate.

6. Chatter Tracking Does Not Record Field Changes

Severity: Medium **Requirement:** Chatter should track business-critical field changes such as Status, Assigned Employee, and Condition.

Live Test Result:

Creating an asset added the standard creation message, but changing a tracked field did not add a new tracking message or tracking values.

Test evidence:

```
before_count 1
after_count 1
tracking_values []
```

The model fields are configured with tracking metadata, but the expected chatter tracking behavior did not occur during the Odoo ORM test.

Recommended Fix:

Investigate why `mail.thread` tracking is not posting tracking values on write for `md.employee.asset`. After the fix, changing `state`, `md_condition`, or `md_employee_id` should create a new chatter message with tracking values.

Requirement Coverage

Requirement	Status	Notes
Create/manage asset records	Pass	Asset model and views exist.
Asset category configuration	Pass	Category model, views, and default data exist.
Assign assets to employees	Pass	<code>md_employee_id</code> , <code>onchange</code> , and employee tab exist.
Track asset status	Pass	<code>state</code> field, <code>statusbar</code> , filters, buttons exist.
Track asset condition	Pass	<code>md_condition</code> field and damaged logic exist.
Employee form Assigned Assets tab	Pass	Added through inherited employee view.
Dedicated menus/actions	Pass with risk	Menus exist; action <code>view_mode</code> should remove search.
List/form/search/kanban views	Pass	All are present.
Default filters/groupings	Partial	Filters/group-by exist; default group by Status not applied.
Chatter tracking	Fail	Field tracking metadata exists, but live write did not create tracking messages.
Security group/access rights	Pass	Menus/access restricted to manager group.
Asset code sequence	Pass	Sequence exists and create method uses it.
Asset code unique	Pass	SQL constraint exists.
Asset code mandatory	Partial	Auto-generation exists, but field is not <code>required=True</code> .
Assigned validation	Partial	Constraint exists; button should validate before state change.
Return date validation	Pass	Constraint exists.
Mark returned	Pass	Sets returned and return date.
Mark damaged	Pass	Sets damaged status and condition.
Mark lost	Pass	Sets lost status.

Requirement	Status	Notes
Scrap	Pass	Sets scrapped and does not archive.
Employee asset count	Partial	Computed, but not visible in UI.

Recommended Functional Test Cases In Odoo 18

Run these after applying the fixes:

1. Install the module on a clean Odoo 18 database.
2. Open Employees > Asset Register > Assets.
3. Create an asset without entering Asset Code and confirm AST-00001 is generated.
4. Try duplicate Asset Code and confirm unique validation.
5. Click Mark Assigned without employee and confirm a validation error appears.
6. Select employee and confirm Assigned Date auto-fills and status becomes Assigned.
7. Click Mark Returned and confirm Return Date auto-fills.
8. Set Return Date earlier than Assigned Date and confirm validation error.
9. Click Mark Damaged and confirm Condition becomes Damaged.
10. Click Mark Lost and Scrap and confirm statuses update correctly.
11. Open the employee form and confirm the Assigned Assets tab shows linked assets.
12. Confirm search filters and group-by options work.
13. Confirm chatter creates tracking messages for status/condition/employee changes.
14. Confirm the asset action opens without errors after removing search from view_mode.

Functional Acceptance Decision

Status: Not fully accepted yet.

The module is mostly aligned with the functional requirement, but the developer should fix the medium findings before final acceptance:

- Make Asset Code truly mandatory while preserving sequence auto-generation.
- Validate employee explicitly in `action_mark_assigned()`.
- Remove search from the asset action `view_mode`.
- Fix chatter tracking so field changes create tracking messages.

The low findings can be fixed in the same pass or accepted if the team is comfortable with the current behavior.